MILITARY SPECIFICATION

ENCODER, SHAFT POSITION TO DIGITAL, CONTACT TYPE, ALTITUDE REPORTING; GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for a contact type altitude reporting encoder.

1.2 Classification -

1.2.1 Nomenclature - The nomenclature shall consist of the item name, followed by a type designation (see Table I).

TABLE I

NOMENCLATURE

ITEM NAME	TYPE DESIGNATION					
Encoder, Shaft Position-to-Digital, Contact Type Altitude Reporting	Size	Rotation	Capacity	Logic	Resolution	Modification
1.2.1.1	1.2.1.2	1.2.1.3	1.2.1.4	1.2.1.5	1.2.1.6	1.2.1.7

1.2.1.1 Item name - In all cases the item shall be "Encoder, Shaft Position to Digital, Contact Type, Altitude Reporting."

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1.2.1.2 Size - The first two digits of the type designation shall indicate the maximum diameter of the encoder in tenths of an inch. If the maximum diameter is not a whole number of tenths, the next higher tenth shall be used.

1.2.1.3 Rotation - The direction of shaft rotation for increasing count, as viewed from the shaft end of the encoder, is denoted by the following symbols:

Clockwise rotation - CW
Counterclockwise rotation - CC

1.2.1.4 Capacity - The capacity of the encoder shall be designated by a four digit number, following the size, indicating the maximum design altitude of the encoder in hundreds of feet.

1.2.1.5 Logic - Whether the contact selection is accomplished internally or externally shall be designated by one letter as follows:

Internal Logic - I
External Logic - E

1.2.1.6 Resolution - The resolution of the encoder shall be designated by a two digit number to indicate the number of discrete counts in hundreds per 360° of the input shaft.

1.2.1.7 Modification - The lower case letter "a" following the resolution designation indicates the original or basic issue of an encoder type. Subsequent modifications which affect interchangeability are designated by succeeding letters in the lower case alphabet except that "i", "l", and "o" shall not be used.

1.2.2 Illustration - An encoder bearing the type designation 11CC1267I80a. is one with the following characteristics:

Size - More than 1.00 inches but not more than 1.10 inches. (11)
Rotation - Counterclockwise as viewed from the shaft extension end for increasing count (CC)
Capacity - 126,700 feet (1267)
Logic - internal (I)
Resolution - 8000 feet per turn (80)
Modification - original issue (a)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

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SPECIFICATIONS

Military

MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical; Preparation for Delivery
MIL-W-16878/4	Wire, Electrical, Type E, 200° and 260° C, 600 Volts, (Insulated, High Temperature)
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-B-22205	Bags, Transparent, Flexible, Heat-Sealable, for Packaging Applications

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-463	Definitions and System of Units, Electromagnetic Interference Technology
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment

(Copies of specifications, standards, drawings, and publications required by

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suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

U.S. National Standard for Common System Component Characteristics for the IFF Mark X(SIF)/Air Traffic Control Radar Beacon Systems SIF/ATCRBS.

(Copies of the FAA document may be obtained from the Federal Aviation Agency, Washington, D.C. 20553.)

3. REQUIREMENTS

3.1 Specification sheets - The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of a conflict between this specification and the specification sheet, the specification sheet shall prevail.

3.2 Qualification - The encoders furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids.

3.3 Quality conformance - Each encoder furnished under this specification shall have demonstrated that it fulfills all of the requirements of this specification by satisfactorily performing all of the acceptance tests specified herein and being one of a lot from which representative units have passed the specified sampling tests.

3.4 Materials - Materials used in the manufacture of encoders shall be suitable for the purpose, and shall conform to the applicable Government specification. Materials conforming to contractor's specifications may be used provided that the specifications are approved by the Government and contain provisions for adequate tests. The use of contractor's specifications will not constitute a waiver of Government inspection.

3.4.1 Metals -

3.4.1.1 Metals shall be of a corrosion resistant type or suitably processed to resist corrosion.

3.4.1.2 Dissimilar metals - Dissimilar metals as defined in MIL-STD-454 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion.

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3.4.1.3 Non-magnetic materials - Non-magnetic materials shall be used for all parts except where magnetic properties are essential.

3.4.1.4 Magnesium - Magnesium or magnesium alloys in any form shall not be used.

3.4.1.5 Aluminum - Aluminum alloy parts shall be in accordance with MIL-E-5400.

3.4.1.6 Nonferrous materials - Nonferrous materials shall be used for all parts, except where ferrous materials are essential.

3.4.1.7 Housings - Encoder housings shall be fabricated from uncoated, corrosion-resistant aluminum alloy or corrosion-resistant steel to insure maximum electrical bonding between housing and mating surfaces. This shall be required of encoders to insure compatibility with the electromagnetic interference requirements of 3.5.8.

3.4.1.8 Rotor shaft - Rotor shafts shall be fabricated from corrosion resisting steel, having a hardness equivalent to Rockwell C20-32.

3.4.1.9 Bearings - Corrosion-resisting ball bearings ABEC Class 5 or better of the radial-thrust type shall be used on the encoder input shaft. Higher precision bearings shall be used in applications where bearing runout will cause a shift of one-half count or greater on the least significant digit. This latter requirement shall be observed as the maximum permissible bearing error.

3.4.1.10 Electrical contacts - Under storage and use environments, electrical contact surfaces shall not form chemical combinations resulting in films of higher electrical resistivities than the parent metal.

3.4.1.11 Semiconductors - Silicon semiconductors conforming to MIL-S-19500 shall be used throughout in encoders requiring semiconductor devices. The reverse bias impedance of isolating diodes when used shall be not less than one megohm throughout the range of operating temperatures.

3.4.2 Non-metallic materials -

3.4.2.1 Fungus resistance - Materials which are nutrients for fungi shall not be used.

3.4.2.2 Fumes - Materials used in the construction of encoders shall not liberate corrosive or toxic fumes or vapors in the range of operating temperatures specified.

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3.4.2.3 Combustion - Materials used in the construction of encoders shall not produce or liberate toxic fumes when burned. Materials shall not sustain combustion.

3.4.2.4 Plastics - Plastics shall be of such composition and workmanship to conform with all aspects of this specification in the manufacture of the end item.

3.4.2.4.1 Impregnating and potting compounds - If impregnating compounds, potting compounds, or both are used, then they shall not at any time, either during application or during service life, have any injurious effect upon any of the other materials used in the encoder. Such compounds shall neither crack nor flow throughout the range of operating and storage temperatures specified in the applicable specification sheet.

3.4.2.4.2 Cast or moulded thermosetting materials - When used for insulation, parts fabricated from cast or moulded thermosetting plastic materials having a moisture absorption of 1.04 percent or greater, shall be dried and impregnated with a suitable material after all machining and punching operations have been completed.

3.4.2.5 Protective coatings - All encoder parts shall be suitably protected so as to be resistant to destructive deterioration caused by the corrosive conditions experienced in service. Destructive deterioration shall be construed as being any type of corrosion which could in any manner prevent the encoders from meeting performance requirements of this specification during service life. Where cleaning operations on metal parts are not specified in detail, they shall be in accordance with approved government specifications or commercial practices which will not cause any subsequent corrosion. Any procedure which would cause maloperation or unreliability shall not be used.

3.4.3 Selection of Government documents - Specifications and Standards for necessary commodities and processes not specified herein shall be selected in accordance with MIL-STD-143 except as specified in 3.4 3.1 and 3.4.3.2.

3.4.3.1 Standard parts - Military standard parts shall be used wherever they are suitable for the purpose and shall be identified on the drawings by their Military Standard part numbers.

3.4.3.2 Commercial parts - Commercial parts having suitable properties shall be used when on the date of invitation for bids there are no suitable Military Standard parts. In any case, commercial hardware such as nuts, bolts, washers, etc., having suitable properties may be used, providing that:

(a) They can be replaced by Military Standard parts without alteration to parts or equipment.

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(b) The corresponding Military Standard part numbers are referenced in the parts list and if practicable on the contractor's drawings.

3.4.4 Lubricants - All vendors shall use lubricants consistent with all requirements of the applicable specification sheet (e.g. temperature, life, speed, and environmental conditions)

3.4.5 Soldering - Soldering shall be in accordance with MIL-STD-454.

3.4.6 Terminal lead wire - Electrical wire used for terminal leads shall conform to MIL-W-16878/4.

3.4.7 Internal wiring - Internal wiring shall be in accordance with MIL-E-5400.

3.4.8 Electronic components - Electronic components shall be in accordance with MIL-E-5400.

3.5 Design and construction -

3.5.1 Threaded parts - Screw threads shall conform to MIL-S-7742

3.5.2 Physical characteristics - The following physical characteristics shall be as specified in the applicable specification sheet

(a) Overall dimensions
(b) Mounting dimensions
(c) Shaft dimensions
(d) Weight
(e) Code range
(f) Scan (logic)
(g) Function
(h) Capacity
(i) Resolution
(j) Full count revolutions
(k) Starting torque
(l) Running torque
(m) Operating speed
(n) Slew speed
(o) Radial play
(p) End play
(q) Shaft run-out
(r) Shaft perpendicularity
(s) Operating temperature range
(t) Lead wire size

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3.5.3 Direction of shaft rotation - Direction of the shaft rotation (as noted in accordance with 1.2.1.3) shall be as viewed from the shaft extension end of the encoder. The shaft shall be capable of continuously rotating freely in the clockwise and in the counterclockwise direction.

3.5.4 Interchangeability - All encoders having the same nomenclature shall be mechanically and electrically interchangeable without change to the encoder, existing parts, or associated circuitry.

3.5.5 Code range -

3.5.5.1 Multi-turn encoders - Multi-turn encoders shall consist of 80 discrete counts for 360° rotation of the input shaft. Each count shall be equivalent to 100 feet of altitude change with the code starting at -1200 feet of equivalent altitude. The code shall be capable of an altitude range up to the design altitude with a maximum design altitude of 126,700 feet.

3.5.5.2 Single-turn encoders - Single-turn encoders shall be capable of digitizing altitude from -1200 feet to the design altitude.

3.5.6 Wire lead identification - Wire leads shall be color coded in conformance with Table II. The minimum lead length shall be as specified in the specification sheet. Each lead shall be capable of undergoing the test in accordance with 4.4.4 without insulation damage, breakage of strands, or lead pullout.

TABLE II

IDENTIFICATION OF LEAD WIRES

ENCODER LEAD	COLOR
Common	White-blue tracer
D_2	White-yellow tracer
D_4	White-brown tracer
A_1	White
A_2	Grey
A_4	Blue

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TABLE II (Continued)

ENCODER LEAD	COLOR
B_1	Green
B_2	Yellow
B_4	Orange
C_1	Red
C_2	Brown
C_4	Black
Active filter power lead if necessary	White-green tracer
Case ground if necessary	White-purple tracer

3.5.7 Electrical characteristics - The following electrical characteristics shall be as specified herein and in the applicable specification sheet.

3.5.7.1 Electrical output - The electrical output of the encoder shall be a parallel unambiguous coded signal in accordance with that specified in the U. S. National Standard for common system component characteristics for the IFF Mark X (SIF)/Air Traffic Control Radar Beacon System SIF/ATCRBS.

3.5.7.2 Diode isolation - Isolation diodes shall be provided integral with the encoder package. The maximum voltage drop across each isolation diode shall not exceed 1 volt. The isolation diodes shall be wired to prevent the flow of current if the high side of a DC voltage source is connected to the encoder common lead and the low side is connected to any other encoder lead.

3.5.8 Electromagnetic interference - Encoders and all associated external logic circuitry shall meet the electromagnetic interference requirements of MIL-STD-461, Class 1D, when tested in accordance with 4.4.17. The definitions and system of units shall be in accordance with MIL-STD-463.

3.5.9 Encoder resistance - When tested in accordance with 4.4.13 and with the input shaft stationary, the effective ("on" state) and minimum ("off" state) encoder resistance shall comply with the resistance values of Table III.

3.5.10 Insulation resistance - When tested in accordance with 4.4.14, the resistance between any lead wire except ground and the encoder case shall not be less than 10 megohms.

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TABLE III

ENCODER RESISTANCE

CONTACT STATE	RESISTANCE (ohms)	CURRENT (Ma)	TEMPERATURE (°C)
On (1)	2000 (max)	1.0	Room
On (1)	1500 (max)	2.5	Room
On (1)	2300 (max)	1.0	-54° C
On (1)	1700 (max)	2.5	-54° C
Off (0)	100 K (min)	0 (open circuit)	Full temperature range

3.5.11 Dielectric strength - Encoders shall display no evidence of internal arcing or flashover and insulation breakdown when tested in accordance with 4.4.15. The leakage current shall not exceed 1 peak milliampere.

3.6 Performance -

3.6.1 Output signal -

3.6.1.1 Electrical noise - When tested in accordance with 4.4.11.1, the signal amplitude noise shall not exceed 10% of the signal amplitude. Edge noise associated with the count transition points shall not extend beyond each transition point by an angular distance greater than that equivalent to ±10 feet of altitude.

3.6.1.2 Transition accuracy - When tested in accordance with 4.4.12, the transition accuracy of the encoder shall be ±10 feet from the theoretically correct altitude referenced to the zero position. For multi-turn encoders, as specified in 3.5.5.1, the actual transition points shall occur with ±27 minutes variation from theoretically correct angle. For single-turn encoders, as specified in 3.5.5.2, the angular variation shall be based on the design altitude.

3.6.2 Altitude modulation - When tested in accordance with 4.4.18, there shall be no degradation of the encoder output signal. Following the test of 4.4.18, the encoder shall also meet the requirements of 3.6.1.1.

3.6.3 Encoder non-activity - Following the thirty days non-activity period specified in 4.4.19, the encoder shall be subjected to and meet the requirements of 3.6.1.1 and 3.6.1.2.

3.7 Environmental -

3.7.1 Temperature altitude - Encoders shall be capable of operation from -1000 to +80,000 feet in combination with any temperature from -55 to +85° C.

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All encoders shall meet the requirements of 3.7.1 when tested in accordance with 4.4.20.

3.7.2 Vibration - Encoders shall be capable of withstanding vibration of frequencies up to 500 cycles per second at an acceleration of 10G without damage or loosening of parts or subsequent degradation of performance. All encoders shall meet the requirements of 3.6.1.1 when tested in accordance with 4.4.16.

3.7.3 Humidity - Encoders shall be capable of operation in an atmosphere of high relative humidity. After testing in accordance with 4 4 21, the encoder shall meet the requirements of 3.5.2 (starting and running torque only) and 3.6.1. There shall be no evidence of corrosion or deterioration of the encoder as a result of this test.

3.7.4 Sand and dust - Encoders shall be capable of operation in the presence of a sand and dust laden atmosphere. After testing in accordance with 4.4.22, the encoder shall meet the requirements of 3.5.2 (starting and running torque) and 3.6.1. There shall be no deterioration of the encoder as a result of this test.

3.7.5 Fungus - Encoders shall be capable of operation after exposure to fungi. After testing in accordance with 4.4.23 the encoder shall meet the requirements of 3.6.1. There shall be no deterioration of the encoder as a result of this test.

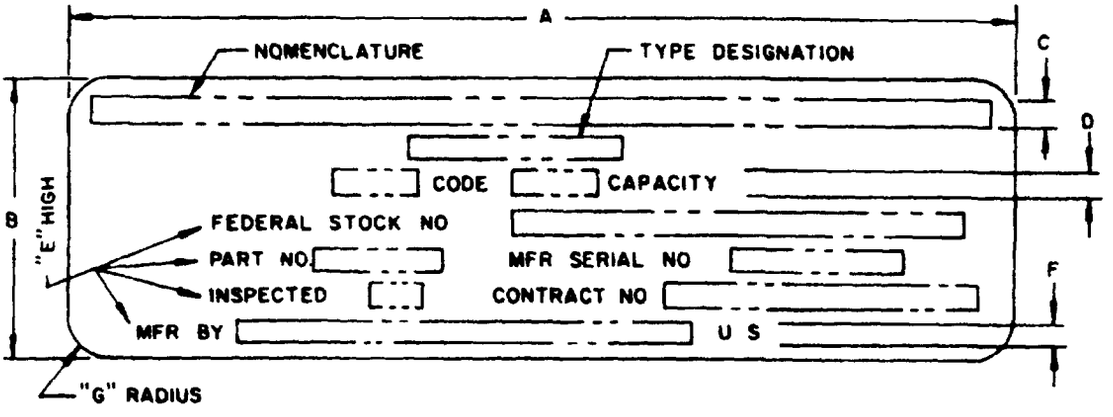
3.7.6 Salt fog - Encoders shall be capable of withstanding atmospheres with salt-laden moisture. After testing in accordance with 4.4.24, the encoder shall meet the requirements of 3.5.2 (starting and running torque) and 3.6.1. There shall be no evidence of corrosion or deterioration of the encoder as a result of this test.

3.7.7 Acceleration - Encoders shall be capable of withstanding structural deterioration caused by acceleration. After testing in accordance with 4.4.25, the encoder shall meet the requirements of 3.6.1. There shall be no evidence of deterioration of the encoder as a result of this test.

3.7.8 Mechanical shock - Encoders shall be capable of withstanding shock blows at an acceleration of 20G of 10 milliseconds duration After testing in accordance with 4.4.26, the encoder shall meet the requirements of 3.6.1.

3.8 Identification - Encoders shall be identified with a nameplate secured in the location specified in the specification sheet. Markings shall conform to MIL-N-18307 and to Figure 1. Identification shall withstand all of the environmental tests required of the encoder.

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TABULATION OF LETTERED DIMENSIONS (INCHES, MAX)

SIZE	A	B	C	D	E	F	G
11, 15	3	3/4	3/32	5/64	3/64	1/16	1/8

NOTES:

1. Legend shall be centrally located horizontally and vertically.
2. All characters shall be gothic style except the modification letter, which shall be lower case elite.
3. Identification marking information shall be as specified in the specification sheet.
4. Manufacturer's part number shall be identical with manufacturer's production drawing number including applicable dash numbers. Changes in manufacturer's part numbers shall be governed by the requirements of MIL-D-1000 regarding drawing numbers.
5. Serial numbers shall be requested from the procuring activity.
6. "Inspected" block requires Government inspector's acceptance stamp.
7. Contract number shall be applied when required by procuring activity.

Figure 1. Identification Marking

3.9 Workmanship - Workmanship shall be in accordance with MIL-STD-454, Requirement 9. In addition, all dimensions and tolerances not specified shall be consistent with the best engineering practice. Where dimensions and tolerances affect the interchangeability, operation, or performance of the encoder they shall be held or limited accordingly.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Test conditions -

4.2.1 Standard test conditions - Unless otherwise specified herein or in the applicable specification sheet, all tests on the encoder shall be performed under the following ambient conditions:

Temperature - +25 ±5° C

Pressure - 28 to 32 inches Hg

Humidity - Room ambient 40 to 90 percent relative humidity

4.3 Classification of tests - The methods of sampling, inspection, and tests conducted on encoders shall be classified as follows:

(a) Qualification

(b) Quality conformance

4.3.1 Qualification inspection - Qualification inspection shall be conducted at a Government laboratory or private test facility approved by the Naval Air Systems Command.

4.3.1.1 Samples - Samples submitted for qualification approval shall be representative of the manufacturer's normal production. The number of samples submitted shall be a minimum of three (3) encoders of each type. Samples of each type for which approval is requested shall be submitted as directed by the qualifying activity.

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4.3.1.2 Test routine - Sample encoders shall be subjected to the qualification tests in the order shown in Table IV

4.3.1.3 Changes - After samples have been submitted for qualification, no changes shall be made in the materials, design, or manufacturing processes without prior approval of the qualifying activity.

TABLE IV

QUALIFICATION TESTS

TEST TITLE	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Visual	4.4 1	
Dimensions	4.4.2	
Weight	4 4 3	
Wire lead stress	4.4.4	
Radial play	4.4.5	
Shaft run-out	4.4.6	
End play	4.4 7	
Shaft perpendicularity	4.4.8	
Concentricity	4.4 9	
Torque	4.4.10	
Starting	4 4.10.1	
Running	4 4 10 2	
Output signal	4 4.11	3.6.1
Electrical noise No 1	4 4.11 1	3.6.1.1
Amplitude noise	4 4.11 3	
Edge noise	4 4.11.4	
Transition accuracy	4.4 12	3.6.1.2
Zero position	4 4.12 1	
Transition accuracy test points	4 4 12 2	
Encoder resistance	4.4 13	3.5.9
Insulation resistance	4.4 14	3.5.10
Dielectric strength	4.4.15	3.5.11
Temperature altitude	4 4.20	3.7.1
Low temperature altitude	4.4.20 1	
High temperature altitude	4.4.20.2	
Vibration error	4.4.16	3.7.2
Electromagnetic interference	4.4.17	3.5.8
Altitude modulation	4.4.18	3.6.2
Encoder non-activity	4.4.19	3.6.3

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TABLE IV (Continued)

TEST TITLE	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Humidity	4.4.21	3.7.3
Sand and dust	4.4.22	3.7.4
Fungus	4.4.23	3.7.5
Salt fog	4.4.24	3.7.6
Acceleration	4.4.25	3.7.7
Mechanical shock	4.4.26	3.7.8
Life	4.4.27	

4.3.2 Quality conformance - Quality conformance tests for acceptance of individual lots shall consist of the tests specified in Tables V or VI.

4.3.2.1 Individual - Unless otherwise specified, each encoder submitted for acceptance shall perform the tests listed in Table V and in the order specified. Failure to meet acceptance criteria shall be cause for rejection.

4.3.2.2 Sampling - Unless otherwise specified, a sample of the manufacturer's production which has passed the individual tests of Table V shall be selected in accordance with MIL-STD-105, Inspection Level II and shall be subjected to the tests listed in Table VI. The acceptable quality level shall be one percent defective for all tests combined.

4.4 Test methods -

4.4.1 Visual inspection - Each encoder shall be inspected for the quality of workmanship. Particular attention shall be given to solder connections, finish and general appearance. Workmanship shall be of the level expected in high quality electrical instruments. The inspection shall include an examination of the identification and connector to assure conformance to the specification. Any evidence of inferior workmanship or disagreement with the specification shall be cause for rejection. The material list and processes shall be inspected for conformance to specification requirements.

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TABLE V

INDIVIDUAL TESTS

TEST TITLE	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Visual	4.4.1	
Dimensions	4.4.2	
Weight	4.4.3	
Wire lead stress	4 4.4	
Radial play	4 4 5	
Shaft run-out	4 4.6	
End play	4.4.7	
Shaft perpendicularity	4 4 8	
Concentricity	4 4 9	
Torque	4 4 10	
Starting	4.4.10.1	
Running	4 4.10 2	
Output signal	4 4 11	3.6.1
Electrical noise No. 2	4 4 11 2	3.6.1.1
Amplitude noise	4 4 11.3	
Edge noise	4 4 11 4	
Transition accuracy	4 4 12	3.6.1.2
Zero position	4 4.12.1	
Transition accuracy test points	4.4 12.2	
Encoder resistance	4 4.13	3.5.9
Insulation resistance	4 4 14	3.5.10

TABLE VI

SAMPLING TESTS

TEST TITLE	TEST PARAGRAPH	REQUIREMENT PARAGRAPH
Transition accuracy	4 4.12	3.6.1.2
Zero position	4 4 12.1	
Transition accuracy test points	4 4.12.2	
Temperature altitude	4 4 20	3.7.1
Low temperature altitude	4 4 20 1	
High temperature altitude	4 4 20 2	
Vibration error	4 4.16	3.7.2
Electromagnetic interference	4.4.17	3.5.8
Altitude modulation	4 4.18	3.6.2
Encoder non-activity	4 4 19	3 6.3

If the failure rate for all tests combined is not less than one percent, the sampling tests shall be applied to 100 percent of production, with the exception of the destructive test which shall remain on a sampling basis. One hundred percent inspection shall remain in force until such time that the failure rate for all tests combined drops below one percent, with a reasonable expectation of remaining below that figure. Failure to meet acceptance criteria shall be cause for lot rejection in accordance with MIL-STD-105

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4.4.2 Dimensions - Each encoder shall have all dimensions that affect interchangeability measured to assure that the dimensions are within the tolerances specified in the applicable specification sheet. Any out of tolerance dimension shall be cause for rejection.

4 4.3 Weight - The encoder including cable and connector shall be weighed on a balance of sufficient accuracy to obtain a value within one percent of the true weight. The weight shall not exceed the value listed on the applicable specification sheet by more than one percent of the listed value.

4.4.4 Wire lead stress- Each wire lead shall be tested by having a two pound weight attached to the extreme end. The weight shall be applied pulling straight away from the encoder, then the lead bent 90° at the point of exit from the housing and while at this attitude, the encoder rotated 360° counterclockwise. Any indication of the lead pulling loose, wire strands breaking, or a permanent deformation of the insulation shall be cause for rejection.

4.4.5 Radial play - The encoder housing shall be rigidly mounted. A dial indicator shall be rigidly mounted in a position to measure shaft movement perpendicular to the shaft axis. The indicator plunger shall contact the shaft as close to the bearing as possible. A force, specified in the specification sheet, shall be applied perpendicular with shaft axis, and within 1/8 inch of the end of the shaft; first in one direction and then in the opposite direction. The difference between the two indicator readings shall be within the range specified in the applicable specification sheet

4 4.6 Shaft run-out - The encoder housing shall be rigidly mounted. A dial indicator shall be mounted to measure shaft eccentricity within 1/8 inch of the end of the shaft When the shaft is rotated the difference between the extreme indicator readings shall not exceed the value specified in the applicable specification sheet.

4.4.7 End play - The encoder housing shall be rigidly mounted in a horizontal position. A dial indicator shall be rigidly mounted in a position to measure shaft motion along the encoder shaft axis. An axial force, specified in the specification sheet shall be applied to the end of the shaft. Note the reading of the dial indicator. Reverse the applied force and note the dial indicator reading. The difference between the two readings shall be within the limits for end play specified in the applicable specification sheet. The encoder shall be energized for this test and no change from the normal code output shall occur at any altitude points. A minimum of ten different altitude points, spaced over the full range of the encoder, shall be checked and recorded.

4 4.8 Shaft perpendicularity - The encoder shaft shall be rigidly mounted in a vertical position. A dial indicator shall be mounted in a position to measure the run-out as close to the outer edge of the mounting surface as possible. When the housing is rotated, the difference between the extreme indicator readings shall not exceed the value specified in the applicable specification sheet.

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4.4.9 Mounting boss concentricity - The encoder shaft shall be rigidly mounted in a vertical position. A dial indicator shall be rigidly mounted in a position to measure the run-out of the periphery of the mounting boss. When the housing is rotated, the difference between the extreme indicator readings shall not exceed the value specified in the applicable specification sheet

4.4.10 Torque - Torque tests shall be performed on a complete encoder assembly in operating condition. The measured torque shall include the torque required to overcome the friction in bearings, contacts, and internal gear train, if used; and the moments of inertia of all rotating parts.

4.4.10.1 Starting torque - The torque required to start the encoder from rest shall be measured in the clockwise and counter-clockwise directions. Starting torque shall not exceed the value specified in the specification sheet.

4 4.10.2 Running torque - The torque required to drive the encoder at a constant angular velocity for at least three revolutions in the clockwise and counter-clockwise directions shall be measured. Running torque shall not exceed the value specified in the specification sheet

4 4.11 Output signal -

4.4.11.1 Electrical noise No. 1 The electrical noise test No. 1 for qualification (4.3.1), and quality conformance sampling plan (4.3.2.2) shall be conducted in the following procedure: The encoder shall be energized with 19 VDC between each lead and common lead. The closed contact current shall be maintained at 2.5 ma. per bit. The encoder shaft shall be rotated at a rate equivalent to 500 feet per minute rate of climb to the maximum altitude capability of the encoder. The encoder output shall be monitored on all bits by a recording oscillograph having a minimum frequency response of 1000 CPS for each channel. A suitable error detection device may be substituted for the recording oscillograph. The test shall be repeated for full-scale descent at 500 feet per minute. A lightbank type readout device is not considered a suitable substitute for a recording oscillograph.

4.4.11.2 Electrical noise No. 2 - The electrical noise test No. 2 for quality conformance individual plan (4.3.2.1) shall be conducted in the following procedure: The encoder shall be energized with 19 VDC between each lead and common lead. The closed contact current shall be maintained at 2.5 ma per bit The encoder shaft shall be rotated at a rate equivalent to 6000 feet per minute rate of climb up to an altitude of 36,000 feet The rate of climb shall then be increased to 24,000 feet per minute rate of climb up to the maximum design altitude of the encoder. The encoder output shall be monitored on all bits by a recording oscillograph having a minimum frequency of 1000 CPS for each channel. A suitable error detection device may be substituted for the recording oscillograph. A lightbank type readout device is not considered a suitable substitute for a recording oscillograph. The test shall be repeated for full-scale descent by rotating the shaft at a

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rate equivalent to 24,000 feet per minute rate of descent from the maximum design altitude to 36,000 feet. The rate of descent shall then be decreased to 6000 feet per minute rate of descent from an altitude of 36,000 feet to sea level.

4 4.11.3 Amplitude noise - The signal amplitude noise shall not exceed 10% of the signal amplitude as measured on the recording oscillograph.

4.4.11.4 Edge noise - The edge noise at transition shall not exceed ±10 feet from the center of the edge noise The center of the edge noise shall not differ by more than 10 feet from the theoretically correct transition altitude.

4.4.12 Transition accuracy - The encoder shall be electrically connected to a suitable encoder readout device and mechanically connected to a suitable dividing head having an accuracy capability of 30 seconds. The encoder shaft shall be rotated and the transition points as indicated by the readout device shall not differ from the theoretically correct transition points by more than ±10 feet of altitude or the equivalent minutes of shaft rotation.

4.4 12.1 Zero position - The mean angular position of the transitions from 0 to -100 feet and 0 to +100 feet shall be used as the zero reference position. To find the mean position (equivalent to 0 feet altitude) turn the encoder shaft in a decreasing count direction to the transition from 0 to -100 feet and note the angular position The encoder shaft shall then be rotated in an increasing count direction to the transition from 0 to +100 feet and the angular position shall be noted The average of the two angular positions shall be used as the zero position of the encoder. All altitude transitions shall be referenced to this angle

4 4 12 2 Transition accuracy test points - Transition accuracy shall be determined for the altitude test points specified in column A of Table VII.

TABLE VII

ENCODER PULSE TRANSITION

A	B	C	
THEORETICAL TRANSITION ALTITUDE (FEET)	PULSE	TRANSITION ON - OFF	TRANSITION OFF - ON
-650	C_2		X
-450	C_1	X	
-250	B_2		X
-50	C_4	X	

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TABLE VII (Continued)

A	B	C	
THEORETICAL TRANSITION ALTITUDE (FEET)	PULSE	TRANSITION ON - OFF	TRANSITION OFF - ON
+50	C_1		X
+350	C_2		X
+650	C_2	X	
+750	B_1		X
+1550	C_4		X
+1750	B_2	X	
+1950	C_4	X	
+2350	C_2		X
+2750	A_4		X
+3750	B_2		X
+4750	B_1	X	
+6250	B_4	X	
+6750	A_2		X
+7050	C_1		X
+7450	C_1	X	
+8750	B_1		X
+9150	C_2	X	
+10750	A_4	X	
+14750	A_1		X
+18750	A_4		X

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TABLE VII (Continued)

A	B	C	
THEORETICAL TRANSITION ALTITUDE (FEET)	PULSE	TRANSITION ON - OFF	TRANSITION OFF - ON
+22750	A_2	X	
+25250	B_4		X
+30750	D_4		X
+35950	C_4		X
+38750	A_2		X
+46750	A_1	X	
+62750	D_2		X
+78750	A_1		X
+94700	D_4	X	
+126750	D_2	X	

4.4.13 Encoder resistance - The effective resistance between each lead and common in the "on" state condition shall be determined while the encoder is at room temperature and at -54° C (during low temperature test 4.4.20.1). With the applied voltage between each lead and common regulated at 19 V.D.C. and current maintained at the levels specified in Table III the effective resistance shall not exceed the values specified in Table III. The minimum resistance between each lead and common in the "off" state (open circuit) condition shall be determined at room temperature, during low temperature tests of 4.4.20.1, and during high temperature tests of 4.4.20.2 and shall meet the requirements of Table III.

4.4.14 Insulation resistance - The insulation resistance shall be measured by applying a potential of 200 V.D.C. between any terminal and the encoder case. The insulation resistance measured shall meet the requirements specified in 3.5.10. Care must be taken to avoid the application of test potential across ground wires and components other than insulation. Measurements shall be made between the mutually insulated points. There shall be no breakdown of insulation or any other permanent damage to the encoder as a result of this test.

4.4.15 Dielectric strength - A potential of 200 V.A.C. rms shall be applied between all contact leads excluding common and ground wires tied together and the encoder case for a period of two minutes. There shall be no arcing or flashover, and the dielectric

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material shall not break down. Care shall be taken to avoid the application of test potential across components other than insulating materials.

4.4.16 Vibration error - The encoder shall be subjected to the vibration test in accordance with Method 514 of MIL-STD-810, Table 514-1, IBIC. While the encoder is undergoing the vibration test, the output signal test (4.4.11) shall be conducted and the requirements shall be met. There shall be no evidence of operational degradation during or at the completion of this test.

4.4.17 Electromagnetic interference - The encoder shall be subjected to the electromagnetic interference test in accordance with MIL-STD-461 and MIL-STD-462 and meet the requirements of 3.5.8. During the test the encoder shall be energized with 15 V.D.C. and the enclosed contact circuit current shall be maintained at 2.5 ma. The encoder shaft shall be driven at a rate of 500 feet per minute rate of climb. The test shall also be performed at a 6,000 feet per minute rate of descent.

4.4.18 Altitude modulation - The encoder shall be connected to a suitable mechanical test fixture to simulate changes of altitude of ±50 feet at a frequency of 0.25 CPS. The encoder shall be energized in accordance with 4.4.11. The encoder shaft shall be adjusted to the position equivalent to altitude of 750 feet, 2750 feet, 6750 feet, and 30750 feet for which altitude modulation of ±50 feet about each of these points shall be performed for two hours. The encoder output shall be monitored by a recording oscillograph having a frequency response of 1000 CPS at the start and near the finish of each two-hour test. There shall be no degradation of signal output of the encoder due to this test. The electrical noise test, 4.4.11, shall be performed at the completion of this test.

4.4.19 Encoder non-activity - After the completion of the Altitude modulation test, 4.4.18, the encoder shall not be subjected to further tests nor shall the shaft be rotated for a period of thirty days. After the completion of the thirty days non-activity, and with no rotation of the encoder shaft prior to testing, the encoder shall be subjected to and meet the requirements of 4.4.11 and 4.4.12.

4.4.20 Temperature altitude -

4.4.20.1 Altitude-low temperature test - The encoder and test fixture shall be placed in a suitable test chamber. The chamber temperature shall be reduced to -54 ±2° C. After the temperature has been stabilized for two hours and while still at the low temperature the encoder shall be subjected to and meet the requirements of the electrical noise test 4.4.11, the encoder resistance test 4.4.13 and the torque test 4.4.10. While maintaining the low temperature, the pressure of the chamber shall be reduced to that equivalent to the high operating altitude of the encoder under test. While at this pressure the encoder shall be subjected to and meet the requirements of the electrical noise test 4.4.11.

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4.4.20.2 Altitude-high temperature test - The encoder and test fixture shall be placed in a suitable test chamber. The chamber temperature shall be increased to +85 ±2° C. After the temperature has been stabilized for two hours and while still at the high temperature the encoder shall be subjected to and meet the requirements of the electrical noise test 4.4.11, the encoder resistance test 4.4.13 and the torque test 4.4.10. While maintaining the high temperature, the pressure of the chamber shall be reduced to the high operating altitude of the encoder under test. While at this pressure the encoder shall be subjected to and meet the requirements of the electrical noise test 4.4.11, the encoder resistance test 4.4.13 and the torque test 4.4.10.

4.4.21 Humidity - The encoder shall be subjected to the humidity test in accordance with Method 507 of MIL-STD-810. At the conclusion of this test, the encoder shall be subjected to and meet the requirements of the electrical noise test, 4.4.11, and the torque test, 4.4.10.

4.4.22 Sand and dust - The encoder shall be subjected to the sand and dust test in accordance with Method 510 of MIL-STD-810. At the conclusion of this test the encoder shall be subjected to and meet the requirements of the electrical noise test, 4.4.11, and the torque test, 4.4.10.

4.4.23 Fungi - The encoder shall be subjected to the fungus test in accordance with Method 508 of MIL-STD-810. At the conclusion of this test, the encoder shall be subjected to and meet the requirements of the electrical noise test, 4.4.11.

4.4.24 Salt fog - The encoder shall be subjected to the salt fog test in accordance with Method 509 of MIL-STD-810. Care shall be taken during this test to insure that the encoder is not sprayed directly with salt solution. At the conclusion of this test, the encoder shall be subjected to and meet the requirements of the electrical noise test, 4.4.11, and the torque test, 4.4.10.

4.4.25 Acceleration - The encoder shall be subjected to the acceleration test for aircraft and helicopters in accordance with Method 513 of MIL-STD-810. At the conclusion of this test, the encoder shall be subjected to and meet the electrical noise test, 4.4.11.

4.4.26 Mechanical shock - The encoder shall be subjected to the shock test in accordance with Method 516 Procedure I of MIL-STD-810. At the conclusion of the test, the encoder shall be subjected to and meet the requirements of the electrical noise test, 4.4.11, and transition accuracy test, 4.4.12. There shall be no deterioration of the encoder as a result of this test.

4.4.27 Life - The encoder shall be energized as specified in 4.4.11 or as otherwise specified. Prior to this test, the encoder shall have satisfactorily passed the electrical noise test, 4.4.11 and the transition accuracy test 4.4.12. At each

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200 hour interval and at the conclusion of the life test, the encoder shall be subjected to and meet the requirements of the electrical noise test, 4.4.11, and the transition accuracy test 4.4.12 and the torque test, 4.4.10. The duration of the life test shall be 3000 hours during which time the encoder shaft shall be rotated to complete the requirements of each cycle in accordance with Table VI. Each cycle shall consist of two (2) days of encoder operation and shall be performed in the order and as specified in Table VIII. At each specified altitude indicated in Figure 2, the encoder shaft shall be sinusoidally modulated at ±50 feet at a frequency of 0.25 cycle per second. The rate of climb or descent between altitude points shall be approximately 3000 feet per minute. For the first hour of each two (2) hours of altitude variation about the shaft position equal to an altitude of 0 feet, the encoder shall not be electrically energized. When the encoder altitude range does not include the higher altitude points, other suitable points shall be chosen and substituted. Upon completion of the last programmed altitude point in the two (2) day cycle, the encoder shaft shall be rotated from the minimum to the maximum encoded altitude and returned to minimum at a rate equivalent to 1000 feet per minute climb and descent. This shall be repeated until the two (2) day cycle is completed. In addition to the above requirements, the encoder shall be operated under temperature conditions specified in Table VIII. Time accrued during electrical operation may be considered as part of the time required by this test. This test may be interrupted for normal weekend shutdown.

TABLE VIII

LIFE TEST CYCLES SCHEDULE

PERIOD 1/	NUMBER OF CYCLES 2/	TEMPERATURE CONDITION
1	2	+25° C
2	1	+71° C
3	2	-35° C
4	2	+25° C
5	2	+45° C
6	1	-54° C
7	3	+25° C

1/ Subsequent periods shall be in above sequence.
2/ One cycle is equal to two days operation.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging -

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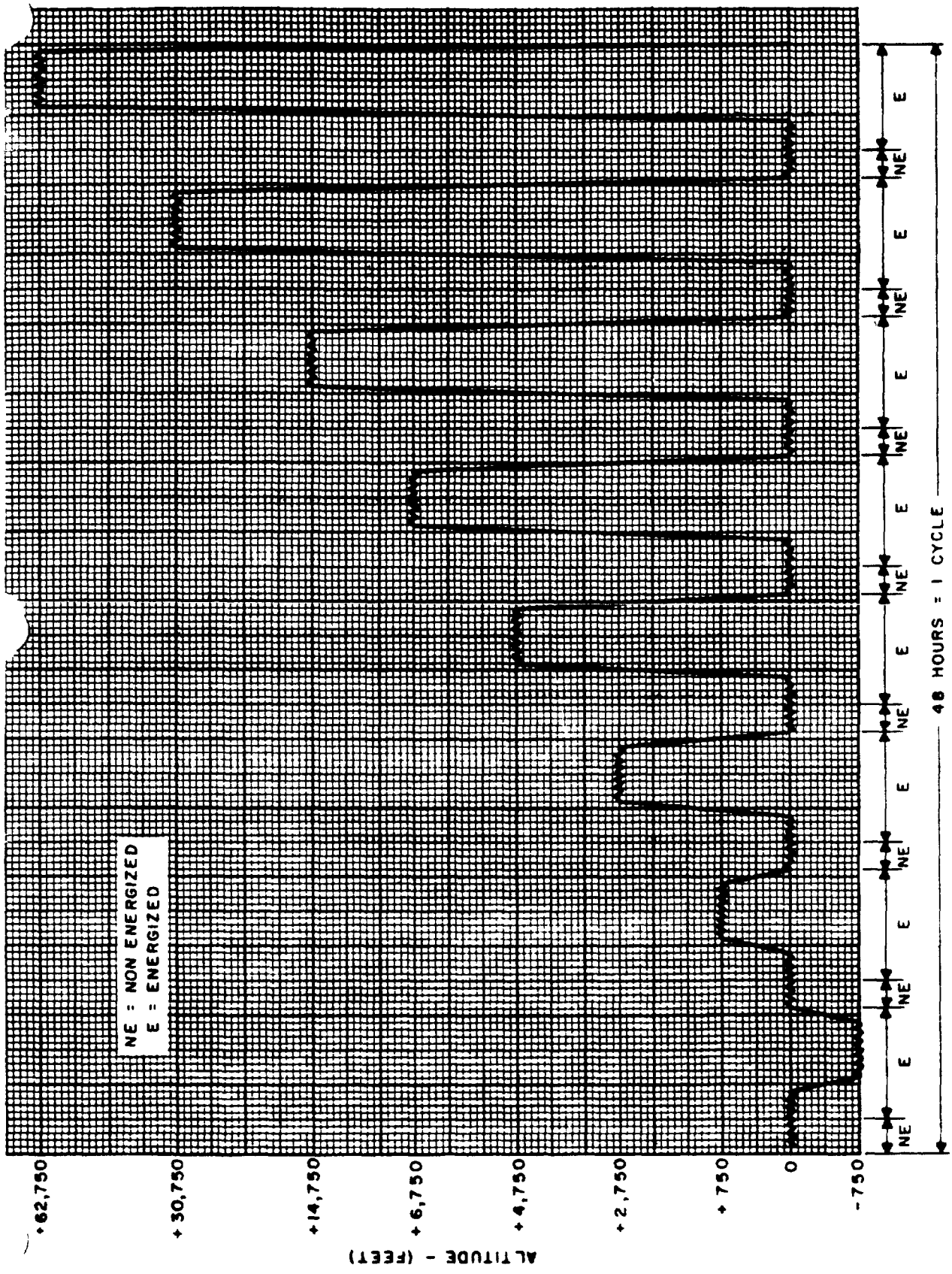


FIGURE 2. ENCODER LIFE TEST

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5.1.1 Cleaning, drying, and preservation - Not applicable.

5.1.2 Unit packaging - Unless otherwise specified by contract or order, each encoder shall be individually packaged in a hermetically sealed container in accordance with MIL-P-7936 with necessary cushioning to provide a tight package The cushioning material shall have a pH value between 6.0 and 8.0 and shall not produce corrosive or deleterious fumes or vapors which could act on or attack the encoder

5 1.3 Hardware - All hardware required by the specification sheet, that is, terminal lugs, terminal screws, shaft washers, and shaft nuts, shall be packaged in a transparent bag (Type III) conforming to MIL-B-22205 The hardware shall be included in the container described in 5.1.2

5.2 Packing - Unless otherwise specified by contract or order, encoders shall be packed in accordance with MIL-P-7936, Level A, B, or C as specified by the procuring activity.

5 3 Marking - The nomenclature for shipment shall be as follows:

Encoder, Altitude Reporting, Contact Type

6 NOTES

6.1 Intended use - The encoders covered by this specification are for use in altitude reporting systems where it is necessary to convert analog data to digital form for display or computation.

6.2 Ordering data - The procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Slant number of specification sheet
(c) Nomenclature of item
(d) Level of packing (see 5.2)

Custodians:
Army - MU
Navy - AS
Air Force -69

Review activities:
Army - MU
Navy - AS
Air Force - 69

User activities:
Army -
Navy -
Air Force -

Preparing activity:
Navy - AS
(Project No 7440-0008)

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Naval Air Engineering Center
Philadelphia, Pennsylvania 19112

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION	
MIL-E-81512	ENCODER, SHAFT POSITION TO DIGITAL, CONTACT TYPE, ALTITUDE REPORTING, GENERAL SPECIFICATION FOR

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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